

File Edit Selection View Go Run Terminal Help python_test

EXPLORER

- PYTHON_TEST
 - DAY-08 Program Outp...
 - DAY-09 Program and ...
 - DAY-10 Program and ...
 - DAY-11 Program and ...
 - DAY-12 Program and ...
 - DAY-13 Program and ...
 - day5_transformations...
 - day6_arithmetic_bitwis...
 - day7.mini.challenges.py
 - day8_color_spaces.py
 - day9_histograms.py
 - day10_smoothing_blu...
 - day11_thresholding.py
 - day13_contours.py
 - day14_mini_project.py
 - dissolve.mp4
 - drawing_basics.py
 - drawing.py
 - edge_preserving_com...
 - enhanced_image.png
 - equalization_comparis...
 - equalize.py
 - equalized_image.png
 - exercise_1_channel_art...
 - exercise_1_contour_an...
 - exercise_1_custom_an...
 - exercise_1_custom_pip...
 - exercise_1_histogram...
 - exercise_1_interactive...
 - exercise_1_threshold...
 - exercise_2_bounding...
 - exercise_2_color_picke...
 - exercise_2_comparison
- OUTLINE
- TIMELINE

exercise_1_custom_analysis.py > ...

```
1 # exercise_1_custom_analysis.py
2 import cv2
3 import numpy as np
4 # Ensure day14_mini_project.py is in the same directory
5 from day14_mini_project import ObjectAnalysisPipeline
6
7 class CustomObjectAnalysis(ObjectAnalysisPipeline):
8     """Extended pipeline with custom geometric analysis metrics"""
9
10    def analyze_contours(self):
11        """Override to add custom shape descriptors safely"""
12        print("\n[Custom] Analyzing contours with additional metrics...")
13
14        # 1. Run parent analysis first to populate results and clean contours
15        super().analyze_contours()
16
17        # 2. Iterate directly over the synchronized data structures
18        # We pair the saved contour arrays with their corresponding result dictionaries
19        for result, contour in zip(self.results['contours'], self.contours):
20            area = result['area']
21            perimeter = result['perimeter']
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
=====
AREA STATISTICS
=====
Min area: 9082
Max area: 9656
Average area: 9369
Total area: 37476
=====
END OF REPORT
=====
PIPELINE COMPLETE!
=====

All outputs saved to: custom_output/
PS C:\Users\raghu\OneDrive\Desktop\python_test>
```

Ln 1, Col 32 (29 selected) Spaces: 4 UTF-8 CRLF {} Python 3.14.5 Python 3.14 (64-bit)

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EXPLORER

- PYTHON_TEST
 - equalization_comparis...
 - equalize.py
 - equalized_image.png
 - exercise_1_channel_art...
 - exercise_1_contour_an...
 - exercise_1_custom_an...
 - exercise_1_custom_pip...
 - exercise_1_histogram...
 - exercise_1_interactive...
 - exercise_1_threshold...
 - exercise_2_bounding...
 - exercise_2_color_picke...
 - exercise_2_compariso...
 - exercise_2_document...
 - exercise_2_equalizatio...
 - exercise_2_noise_redu...
 - exercise_2_object_cou...
 - exercise_3_d_color_hi...
 - exercise_3_color_isolat...
 - exercise_3_crop_objec...
 - exercise_3_custom_ker...
 - exercise_3_multi_objec...
 - exercise_3_size_classifi...
 - exercise_4_backgroun...
 - exercise_4_color_spac...
 - exercise_4_edge_prese...
 - exercise_4_histogram...
 - exercise_4_shape_clas...
 - exercise_5_contrast_en...
 - exercise_5_deblurring...
 - exercise_5_document...
 - exercise1.brighthness.c...
 - exercise1.dav2.py

exercise_3_size_classification.py

```
1 # exercise_3_size_classification.py
2
3 import cv2
4 import numpy as np
5
6
7 def classify_by_size(image_path):
8     """Classify objects by size (small, medium, large)"""
9
10    image = cv2.imread(image_path)
11
12    if image is None:
13        print("Error: Could not load image")
14        return
15
16    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
17    blurred = cv2.GaussianBlur(gray, (5, 5), 0)
18
19    _, binary = cv2.threshold(
20        blurred,
21        0,
```

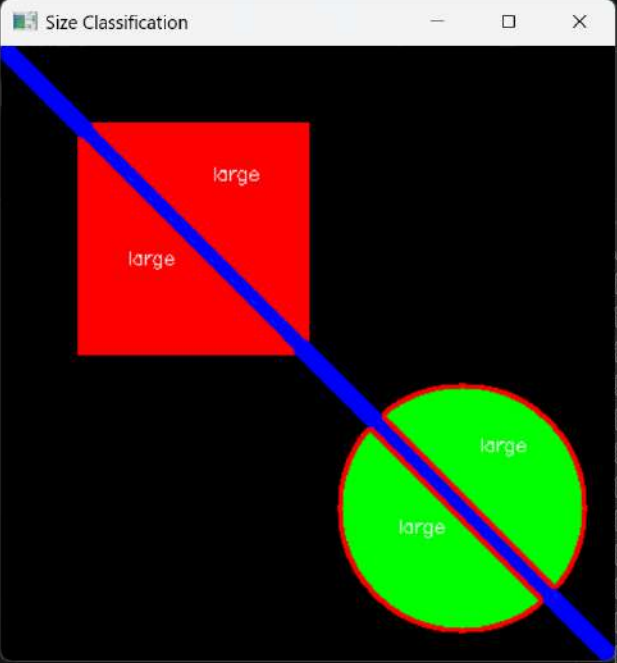
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_3_size_classification.py

Size Classification Results

small: 0 objects
medium: 0 objects
large: 4 objects
[]

Size Classification



python

Ln 1, Col 2 (34 selected) Spaces: 4 UTF-8 CRLF {} Python 3.14.5 Python 3.14 (64-bit)

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EXPLORER

PYTHON_TEST

- edge_preserving_com...
- enhanced_image.png
- equalization_comparis...
- equalize.py
- equalized_image.png
- exercise_1_channel_art...
- exercise_1_contour_an...
- exercise_1_custom_an...
- exercise_1_custom_pip...
- exercise_1_histogram...
- exercise_1_interactive...
- exercise_1_threshold...
- exercise_2_bounding...
- exercise_2_color_picke...
- exercise_2_compariso...
- exercise_2_document...
- exercise_2_equalizatio...
- exercise_2_noise_redu...
- exercise_2_object_cou...
- exercise_3_3d_color_hi...
- exercise_3_color_isolat...
- exercise_3_crop_objec...
- exercise_3_custom_ker...
- exercise_3_multi_objec...
- exercise_3_size_classifi...
- exercise_4_advanced_s...
- exercise_4_backgroun...
- exercise_4_color_spac...
- exercise_4_edge_prese...
- exercise_4_histogram...
- exercise_4_shape_clas...
- exercise_5_complete_a...
- exercise_5_contract an...

exercise_5_complete_app.py

```
1 # exercise_5_complete_app.py
2 import cv2
3 import numpy as np
4 import os
5 from datetime import datetime
6
7 class CompleteImageAnalysisApp:
8     """Complete application with dynamic grid preview matrices"""
9
10    def __init__(self, image_path):
11        self.image = cv2.imread(image_path)
12        if self.image is None:
13            raise ValueError(f"Could not load image from target path: {image_path}")
14
15        self.gray = cv2.cvtColor(self.image, cv2.COLOR_BGR2GRAY)
16        self.original = self.image.copy()
17        self.steps = []
18
19    def add_step(self, name, image):
20        """Add a step to the pipeline"""
21        self.steps.append((name, image))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PROCESSING TIMELINE LOGS

```
=====
Stage 1 : Original
Stage 2 : Grayscale
Stage 3 : Blurred
Stage 4 : Equalized
Stage 5 : Binary
Stage 6 : Edges
Stage 7 : Contours
Stage 8 : Analysis
=====
```

STATUS: COMPLETED | Total Logged Pipeline Blocks: 8

→ Processed panel map successfully written to: 'complete_pipeline.png'

Complete Analysis Pipeline Matrix Layout

Ln 1, Col 29 (26 selected) Spaces: 4 UTF-8 CRLF {} Python 3.14.5 Python 3.14 (64-bit)